

Supplement to Using Publicly Available Data to Predict Outbreak Trends

Warning: package 'knitr' was built under R version 4.5.2

1 Overview

The Supplementary Materials provided in this document include the processing, exploratory, and analysis code, as well as supplementary figures mentioned in the manuscript for the final data analysis project of EPID 8060E S2026, “Using Publicly Available Data to Predict Outbreak Trends”, by Elle Adams.

2 Code and file information

After downloading the BEAM Dashboard data (this analysis used the version that was updated most recently before March 4, 2026, placed in /data/rawdata), the processing.qmd file (/code/processing-code) should be run first, all required libraries are included at the top and there is a description of what each step is for throughout. Processed data was stored in /data/processed-data/processeddata.rds After that, both the exploratory.qmd (/code/exploratory-code) and analysis.qmd (/code/analysis-code) files call this processed code at the beginning of each file and either can be run, again with required libraries at the top and some explanations throughout. Exploratory code includes some more basic figures investigating the dataset and Analysis code includes the all of the model fitting code.

The write up for the manuscript is under /products/manuscript/manuscript.qmd.

The figures in the main manuscript and here in the supplementary can be found under /results/figures.

3 Additional Method Details

Model fitting methods (in the analysis.qmd file) followed the tidymodels structure of first fitting with training data, then doing CV (10 folds), refitting to the model that had the best metric of interest from the CV again with the training data, and finally using the testing data to test the fit of the model on data it has not seen before. Each model, whether regression or classification, followed this general set up and should be reproducible.

Regression models were graphed via scatter plot of predicted outcomes and true outcomes from both test and training data. These are included below.

Classification models were graphed via confusion matrix of predicted outcomes and true outcomes from the testing data. These are included below.

4 Additional results

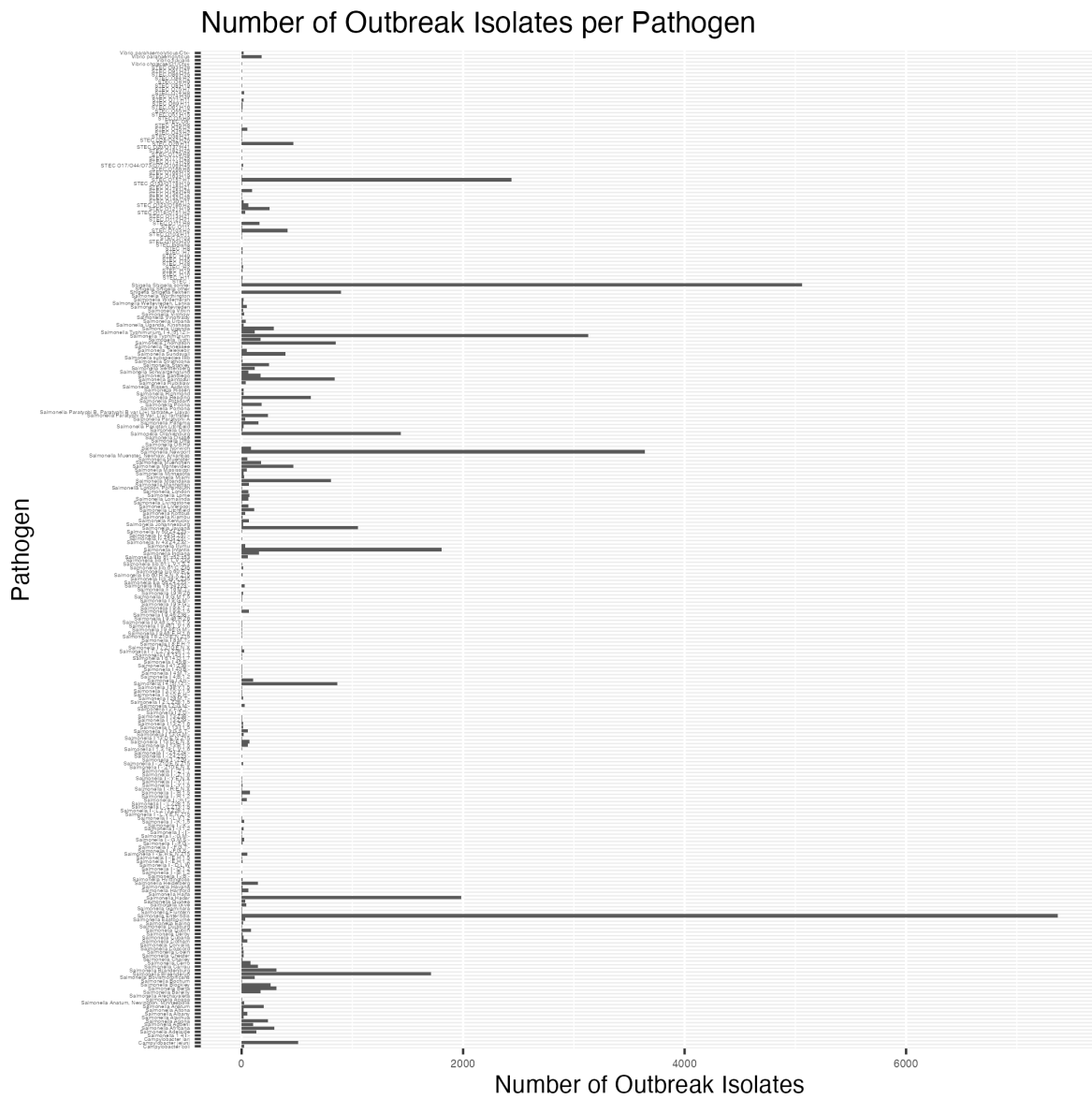


Figure 1: Average Outbreaks by Month

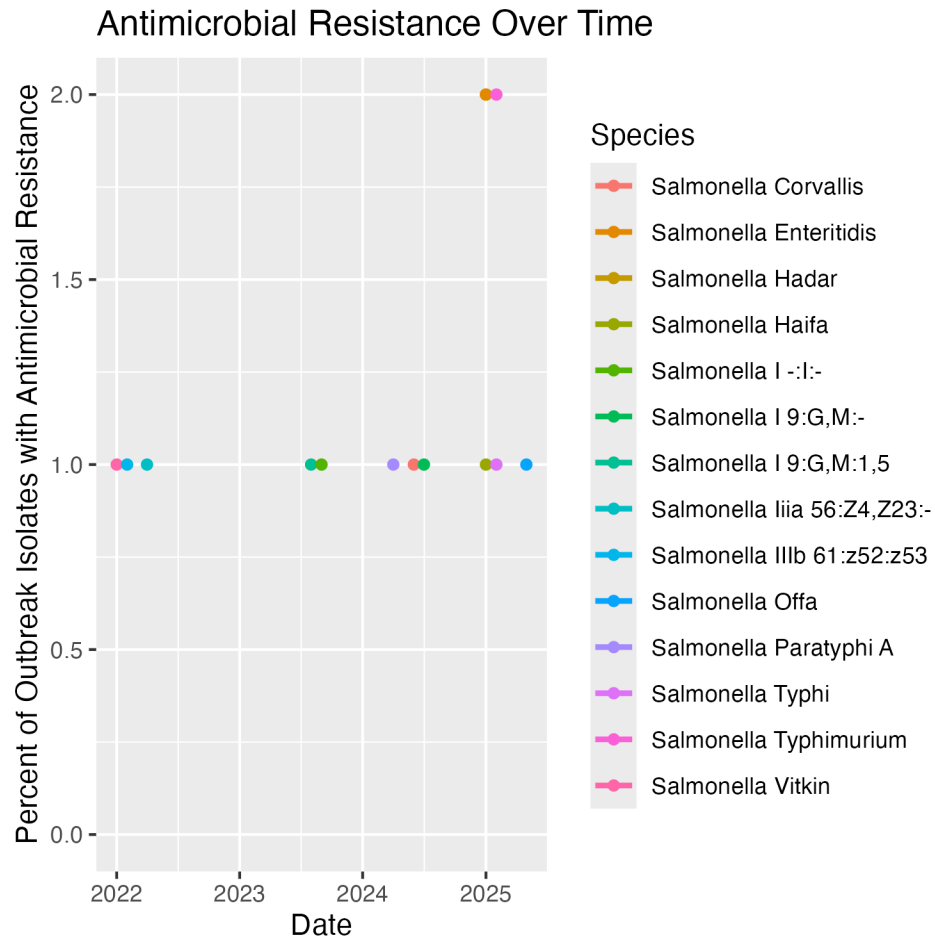


Figure 2: Antimicrobial Resistance per Pathogen Over Time

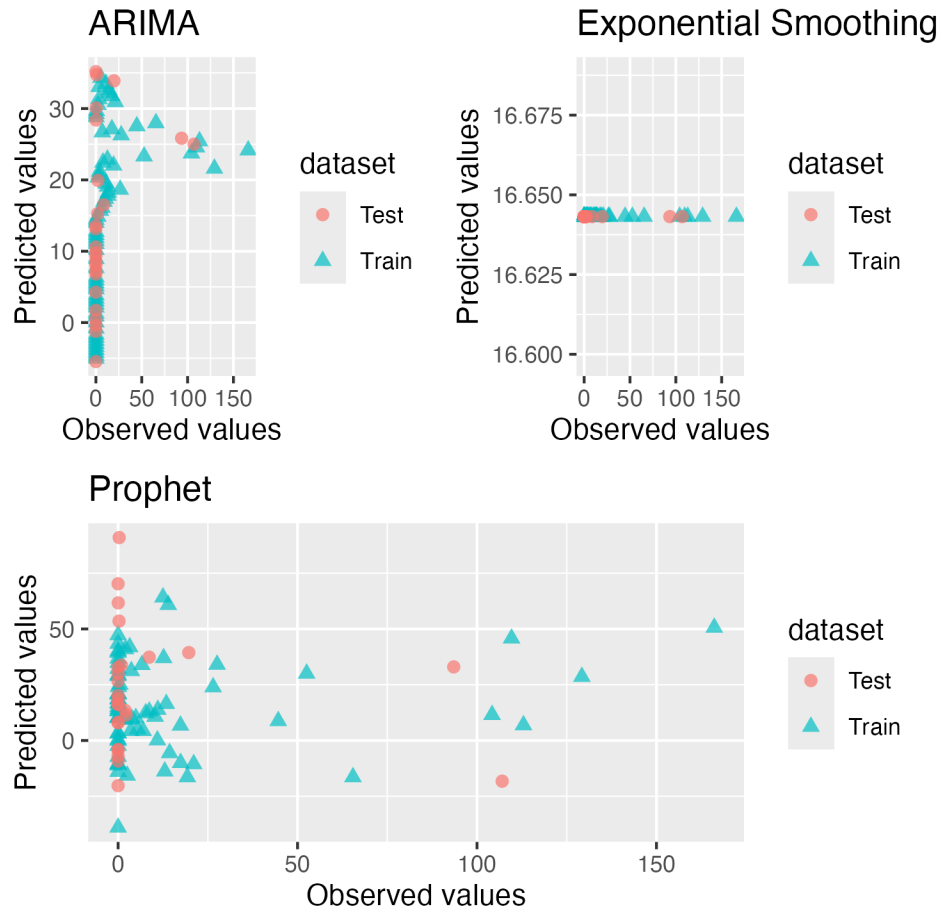


Figure 3: Predicted vs Observed: Outbreaks Over Time

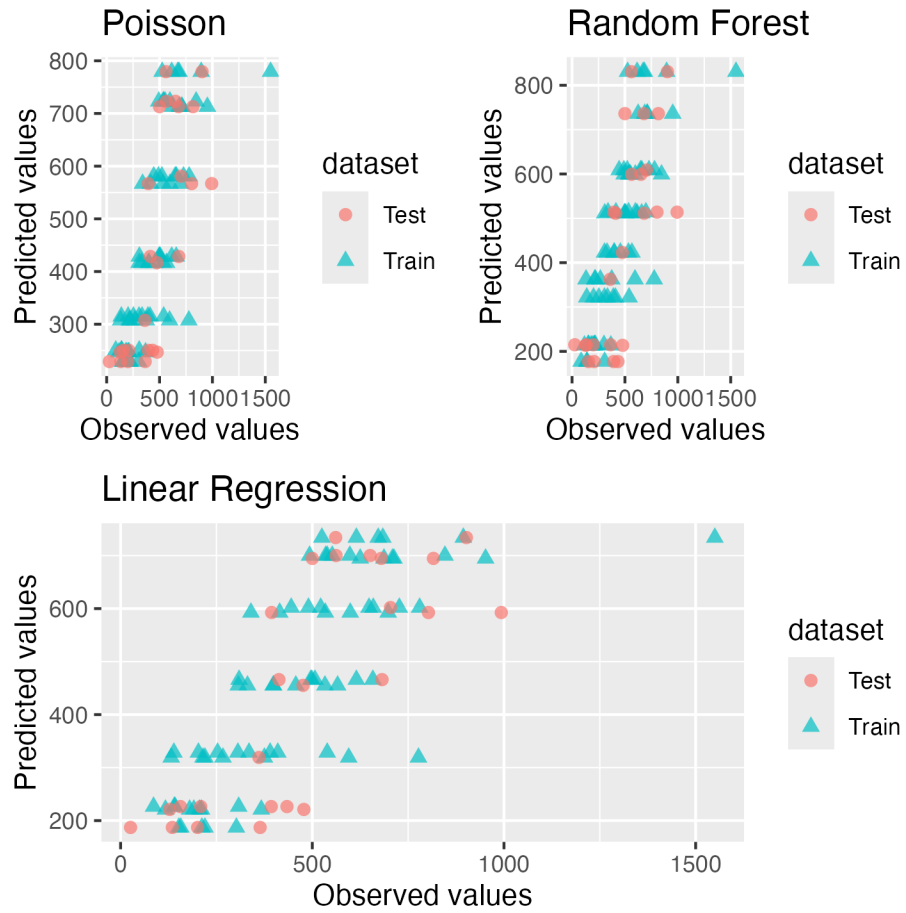


Figure 4: Predicted vs Observed: AMR Over Time

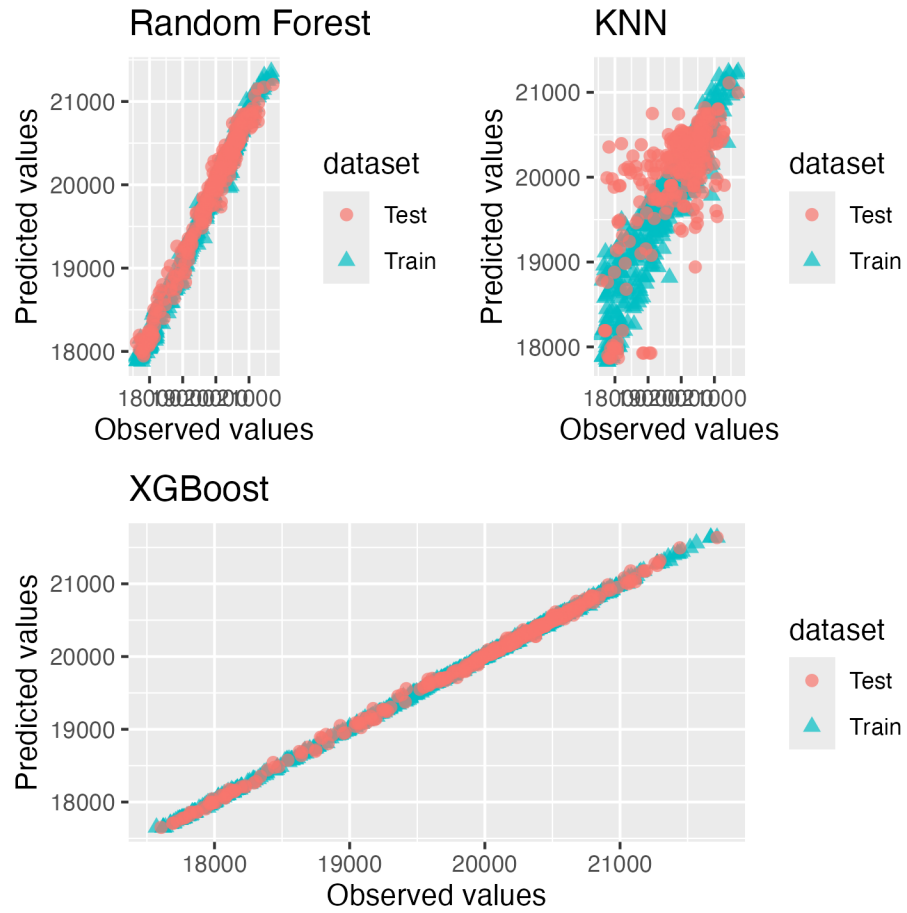


Figure 5: Predicted vs Observed: Next Outbreak: WHEN

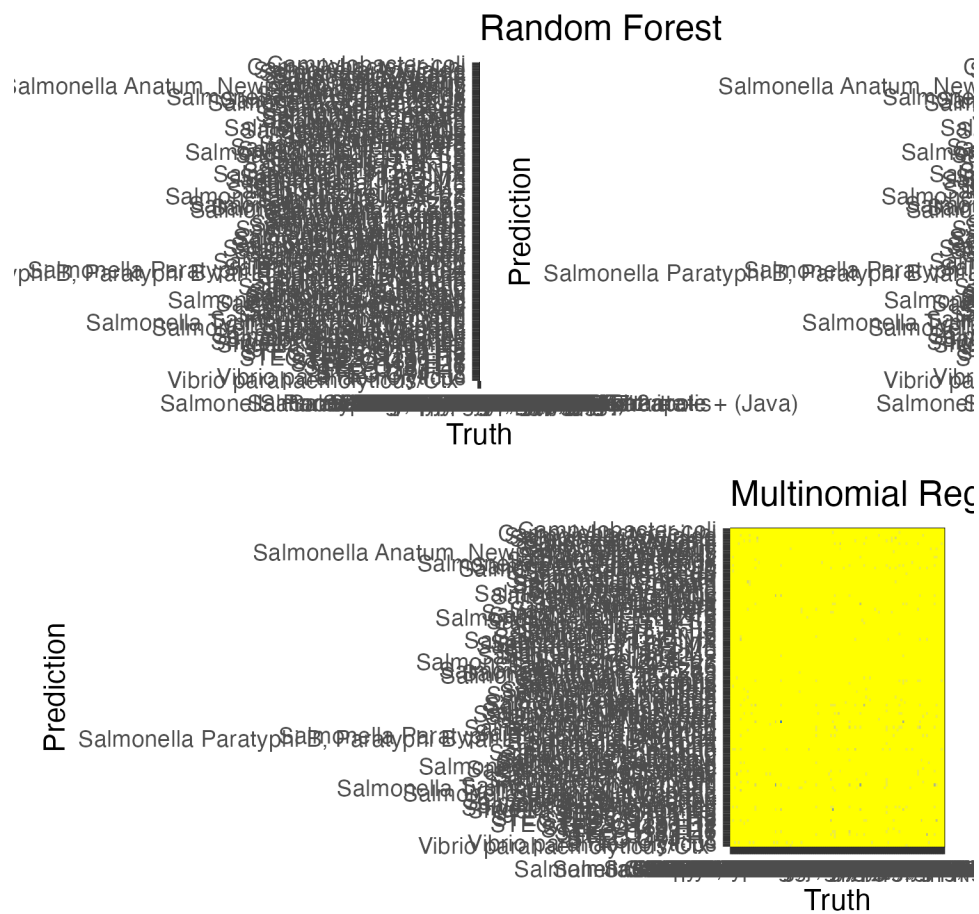


Figure 6: Predicted vs Observed: Next Outbreak: PATHOGEN

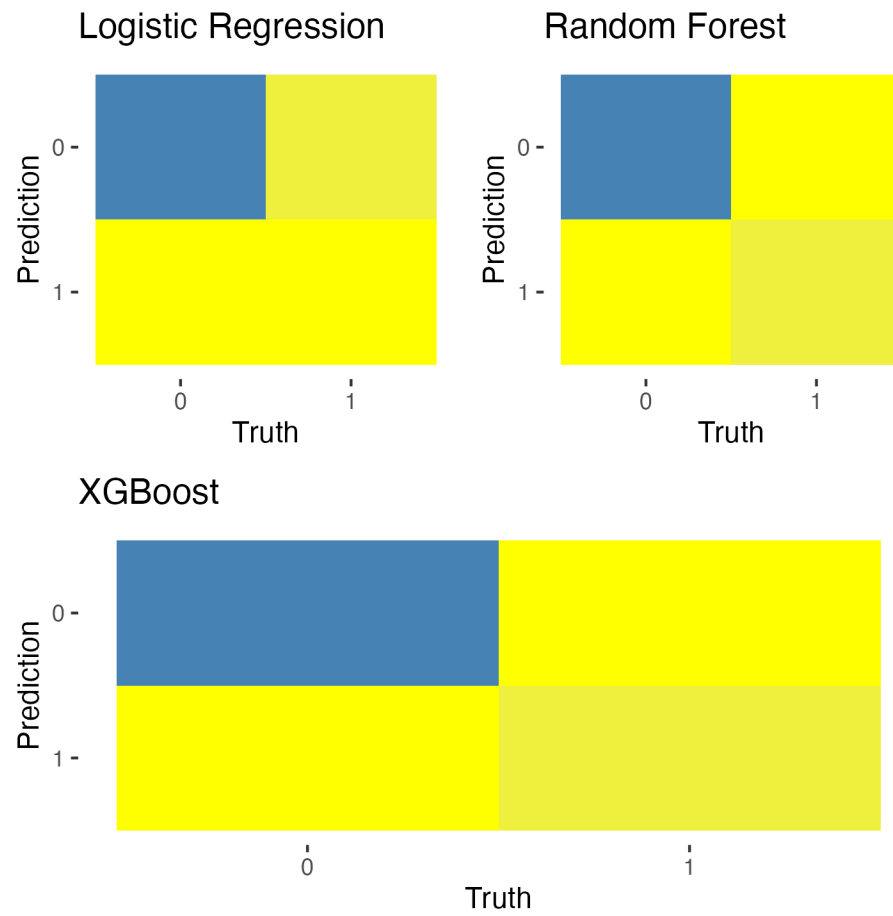


Figure 7: Predicted vs Observed: Next Outbreak: AMR

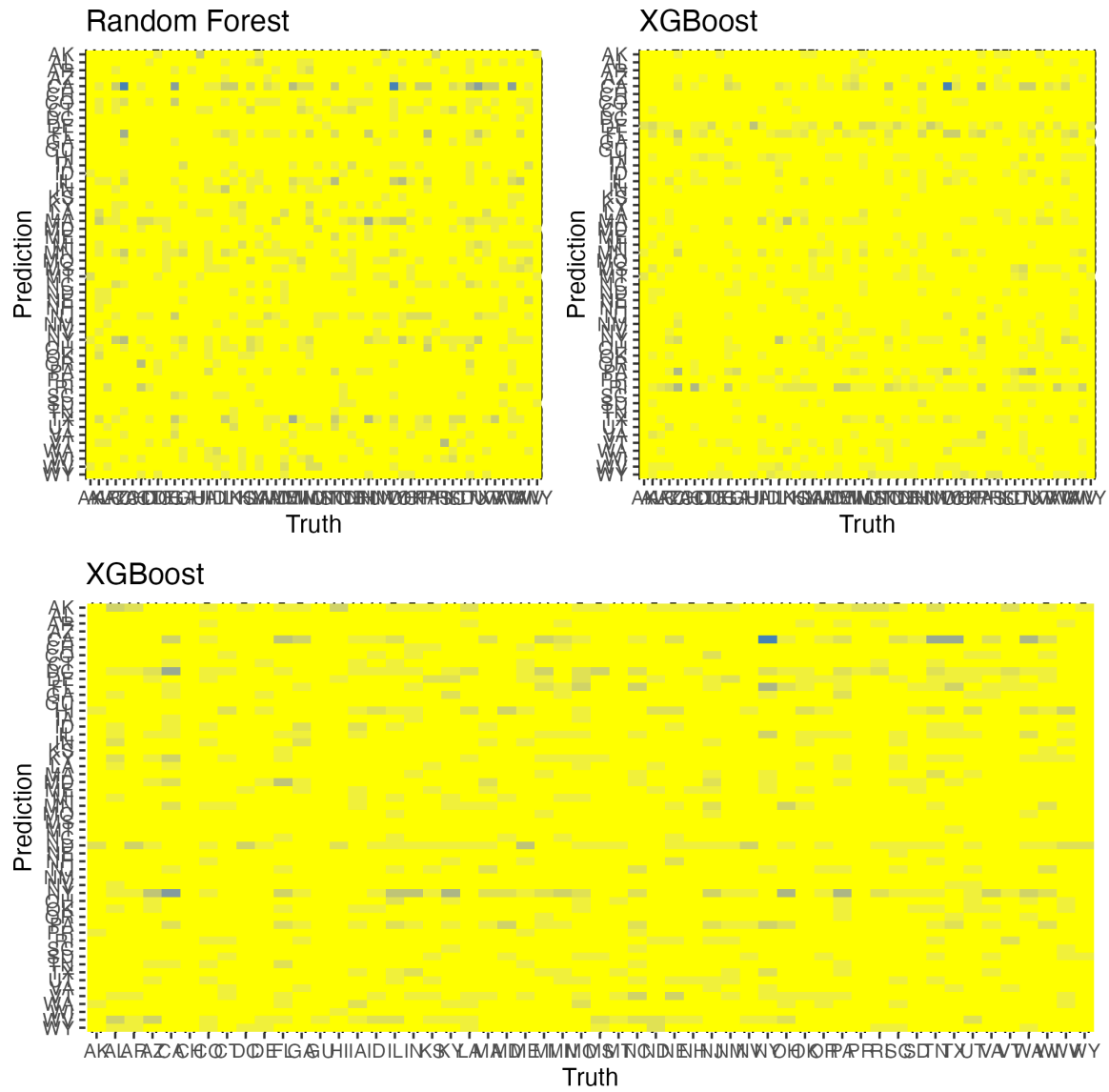


Figure 8: Predicted vs Observed: Next Outbreak: WHERE